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Brain MRI Segmentation

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Development of Advanced Computer Science Applications

Abstract

1. Introduction

2. Data Exploration

2.1 Dataset Introduction

The dataset source comes from "*Brain MRI segmentation*" by Mateusz Buda [1]. The dataset in turn is the LGG Segmentation Dataset, which focuses on the segmentation of abnormal regions in brain MRI images from patients diagnosed with lower-grade glioma (LGG). This dataset was built from medical imaging data obtained from The Cancer Imaging Archive (TCIA) and is linked to cases from The Cancer Genome Atlas (TCGA) lower-grade glioma collection. In addition to imaging data, the dataset also includes clinical and genomic information for each patient through the file data.csv.

This dataset is designed for the study of brain tumor abnormalities visible in FLAIR MRI sequences, and it is commonly used in medical image segmentation and radiogenomic analysis. The segmentation masks were created manually and serve as the ground truth for supervised learning tasks.

The dataset is especially relevant because it combines two important sources of information:

1. Medical imaging data, which supports the segmentation task.
2. Patient-level genomic and clinical data, which can be used for complementary analysis and future radiogenomic studies.

2.2 Structure of the Dataset

The dataset is organized into **110 folders**, each corresponding to one patient. Every folder contains multiple MRI slices and their corresponding segmentation masks. The files follow a naming convention that allows each MRI image to be matched directly with its mask.

The general naming format is:

```
TCGA_<institution-code>_<patient-id>_<slice-number>.tif
```

and the corresponding mask is:

```
TCGA_<institution-code>_<patient-id>_<slice-number>_mask.tif
```

This means that for each MRI slice there is an associated mask file with the same identifier and the suffix _mask.

2.3 File Types and Format

The images are stored in .tif format and each image has 3 channels. These channels correspond to different MRI sequences:

- pre-contrast
- FLAIR
- post-contrast

For some patients, one of these sequences is missing. In those cases, the missing sequence was replaced with the FLAIR sequence so that all images maintain a consistent 3-channel structure.

The segmentation masks are also stored in .tif format, but they are single-channel binary images. These masks indicate whether each pixel belongs to the abnormal region visible in the FLAIR sequence. In the explored dataset, the mask values are represented as 0 for background and 255 for abnormal region, which can be converted into 0 and 1 for training purposes.

2.4 Number of Patients and Samples (instances)

The dataset contains information for 110 patients.
From the exploratory analysis, the dataset includes:

- 3929 MRI images
- 3929 segmentation masks

All MRI images have shape:

- $256 \times 256 \times 3$

All masks have shape:

- 256×256

Thus, the dataset is composed of 2D MRI slices rather than full 3D volumes, and each slice is paired with its corresponding segmentation mask.

2.5 Additional Information

Besides the image and mask files, the dataset includes a file called `data.csv`, which contains patient-level metadata. This file provides information such as:

- Genomic subtype clusters: molecular groupings of tumors based on genomic data, used to identify biologically similar glioma subtypes.
- Histological type: tissue-based classification describing the tumor according to its microscopic characteristics.
- Tumor grade: indicates the severity or aggressiveness of the tumor.
- Tumor location: anatomical region of the brain where the tumor is found.
- Gender: biological sex of the patient.

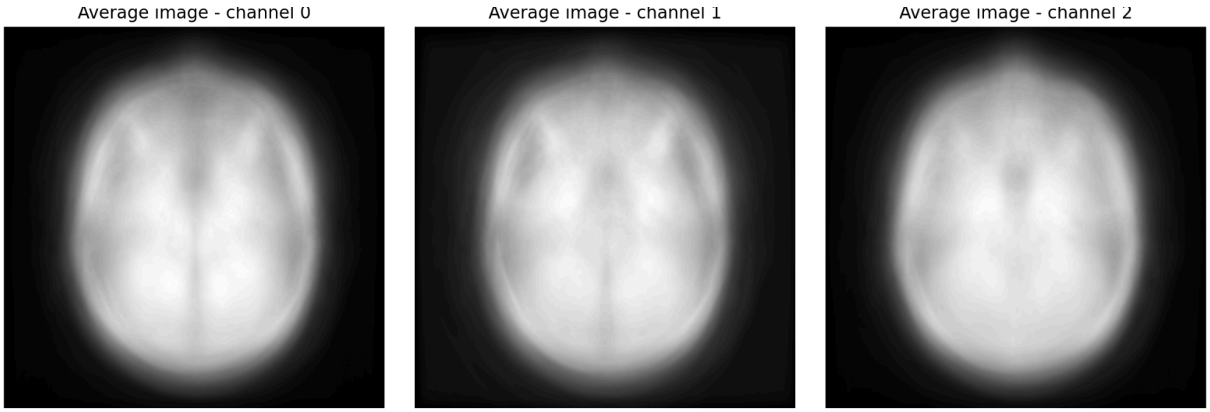
- Age: patient age at the time of diagnosis or pathological evaluation.
- Race: demographic category reported for the patient.
- Ethnicity: ethnic background reported for the patient.
- Survival-related information: variables associated with patient outcome, such as survival status or death indicator.

This metadata is not the direct target of the segmentation problem, but it is useful for understanding the patient and supporting complementary analyses.

2.6 Exploration

2.6.1 Average Image

The following image is a recreation of the average image of the dataset for the three channels.

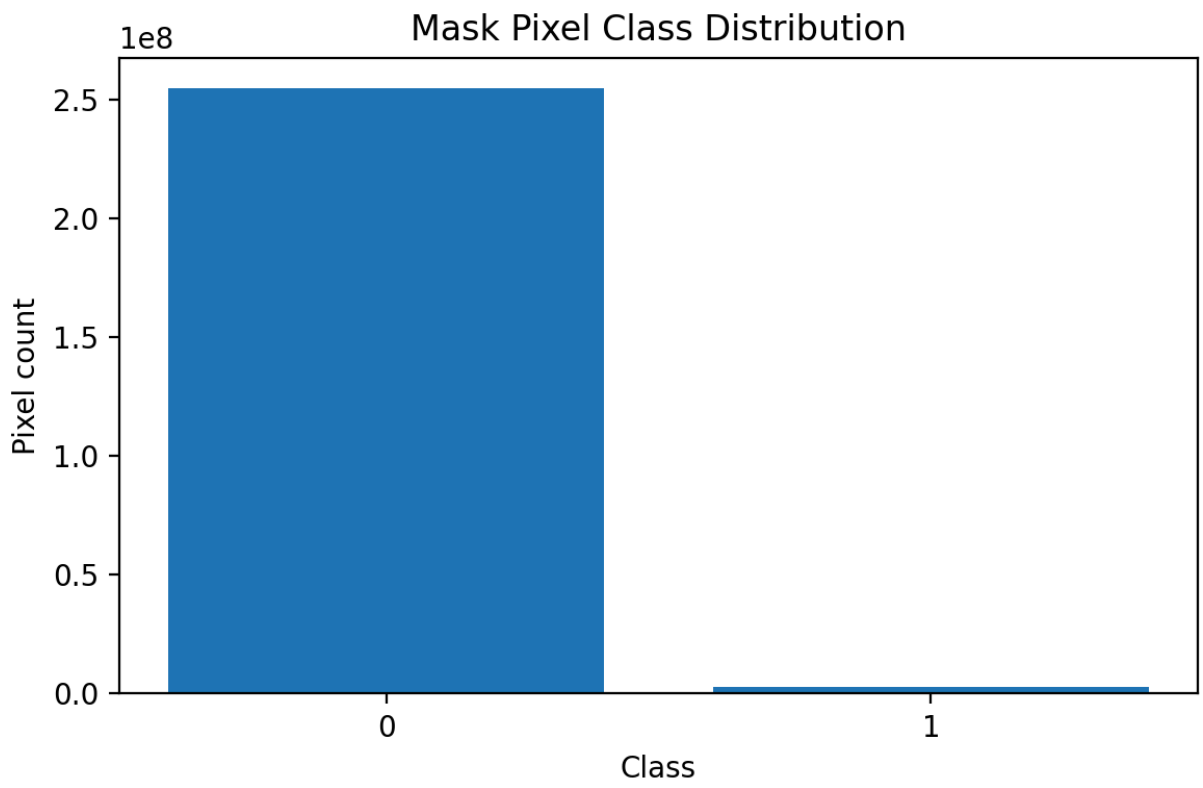


2.6.2 Class distribution

For the image segmentation task, the labels are provided through the segmentation masks, where each pixel belongs to one of two classes.

- **Class 0:** Represents the background, meaning pixels that do not correspond to the abnormal region and therefore are not the target of interest.
- **Class 1:** Represents the FLAIR abnormality region, which is the area that the model is expected to detect and segment.

Pixel class distribution		
Class	Number of pixels	Percentage
0	254839090	98.97%
1	2651854	1.03%



We can note that the dataset is far from being balanced, this due the nature of the dataset and difficult to get new instances.

2.7 Data quality

The quality of the dataset is appropriate for a medical image segmentation task, since it provides a consistent structure across samples and includes manually annotated masks that serve as reliable ground truth. All MRI slices have the same spatial resolution of 256 × 256 pixels, and the images are stored in a uniform 3-channel .tif format, which facilitates preprocessing and model training. In addition, each image has a corresponding segmentation mask, allowing a direct pairing between input data and target labels. However, the dataset also presents some challenges that must be considered during analysis, such as the strong class imbalance.

3.0 Data Preparation

3.1 Data split

To split the dataset, a patient-level strategy was selected in order to avoid data leakage. Since each patient contains multiple MRI slices, all images and masks from the same patient must remain in the same subset. First, a final test set is reserved using approximately 10–15% of the 110 patients. The remaining patients are then used in a 5-fold cross-validation scheme. In each fold, 4 parts are used for training and 1 part is used for validation. This approach allows the model to be validated several times while keeping an independent final test set for the final evaluation.

Stage	Distribution	Approximate patients	Purpose
Final test set	10–15%	11–17	Final unbiased evaluation
Remaining data	85–90%	93–99	Used for cross-validation
Training in each fold	4/5 of remaining data	about 74–79	Model training
Validation in each fold	1/5 of remaining data	about 19–20	Model validation

3.2 Data augmentation

In order to improve the robustness of our data set, the next data augmentation modifications will be applied for the training data, with

Data augmentation	Description / purpose
Horizontal flip	Introduces spatial variation in the image orientation and helps the model generalize better.
Small rotations	Simulates slight changes in image orientation during acquisition while preserving the anatomical structure.
Small shifts / translations	Moves the image slightly in horizontal or vertical directions to improve robustness to position changes.
Small scaling / zoom	Helps the model adapt to small variations in the apparent size of the abnormal region.
Brightness adjustment	Simulates differences in image intensity between scans.
Contrast adjustment	Improves robustness to variations in tissue contrast and scanner conditions.

The selected augmentations are not intended to be applied aggressively. Instead, they are used with small magnitudes and controlled probability. In this way, augmentation increases variability in the training data without introducing unrealistic distortions.

3.3 Prepared dataset

After the data augmentation applied to training data and the separation into folds, the ramming processed data follows the next structure:

Stage / Fold	Subset	Patients	Image-mask pairs
Final split	Train + Validation pool	99	3599
	Final Test	11	330
Fold 1	Train	79	2809
	Validation	20	790
Fold 2	Train	79	2784
	Validation	20	815
Fold 3	Train	79	2926
	Validation	20	673
Fold 4	Train	79	2942
	Validation	20	657
Fold 5	Train	80	2935
	Validation	19	664

Modeling

Evaluation

General Conclusions

Future Work

References

[1] M. Buda, “Brain MRI segmentation,” *Kaggle*, [Online]. Available: https://www.kaggle.com/datasets/mateuszbuda/lgg-mri-segmentation/data. [Accessed: Apr. 12, 2026].

[2] K. Alomar, H. I. Aysel, and X. Cai, “Data augmentation in classification and segmentation: A survey and new strategies,” *Journal of Imaging*, vol. 9, no. 2, p. 46, 2023. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC9966095/. [Accessed: Apr. 12, 2026].